

Lecture 2 Tutorial — Data and Statistics

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Q1. Key Terms (Fill in the blanks)

Complete the statements.

1. A parameter describes a _____.
2. A statistic describes a _____.
3. A sample is a subset of a _____.

Q1 - Solution

Q1. Key Terms (Fill in the blanks)

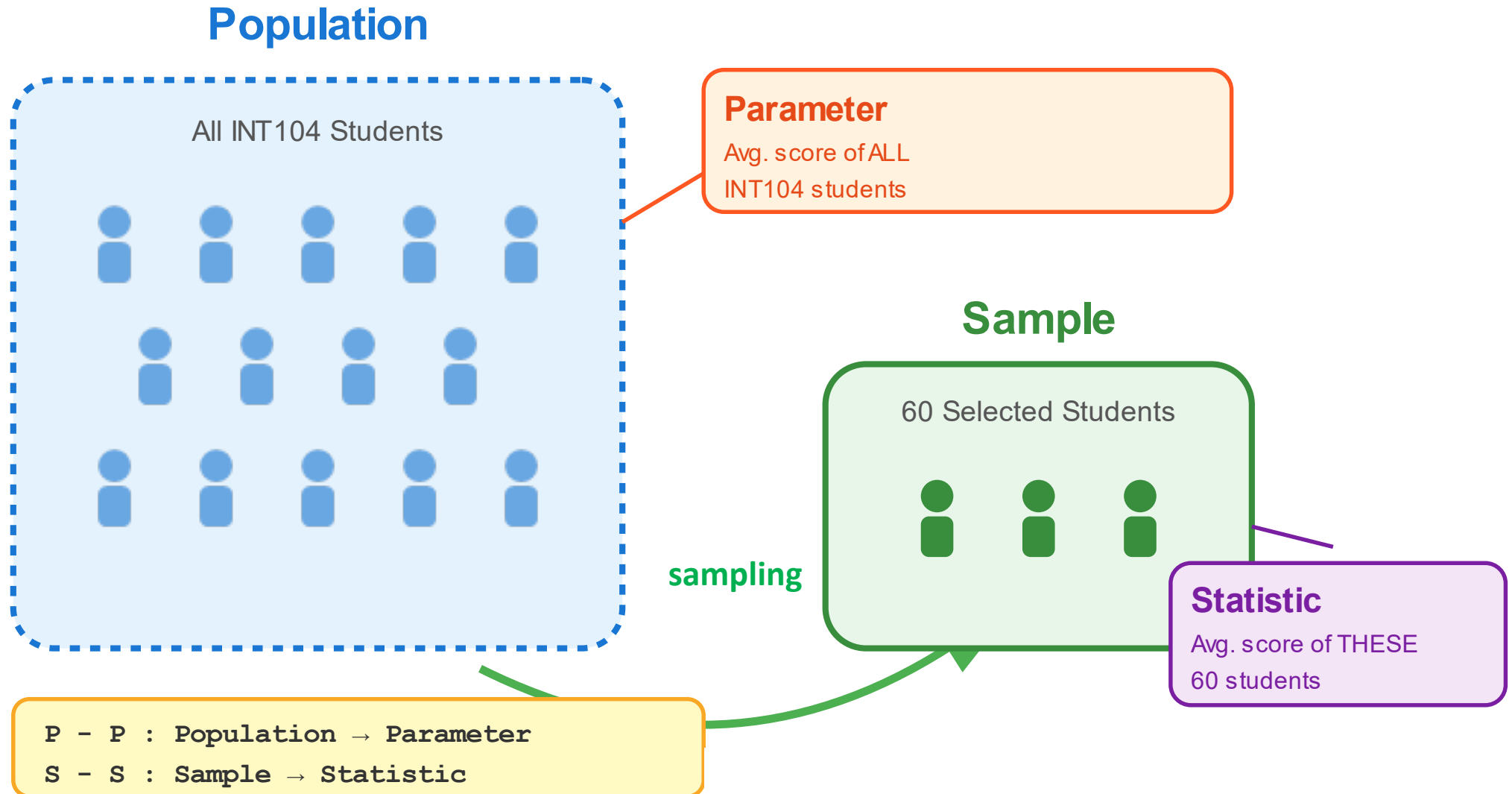
Complete the statements.

1. A parameter describes a _____.
2. A statistic describes a _____.
3. A sample is a subset of a _____.

1. population; 2. sample; 3. population.

Q1

Example

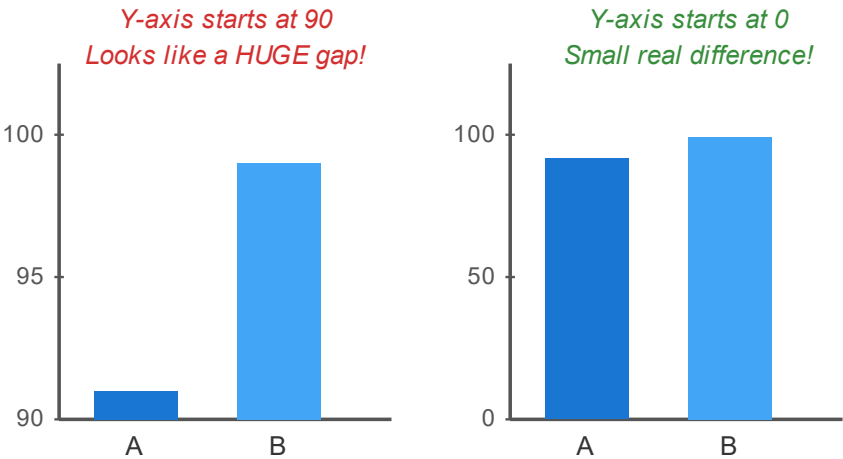


Q2. True / False

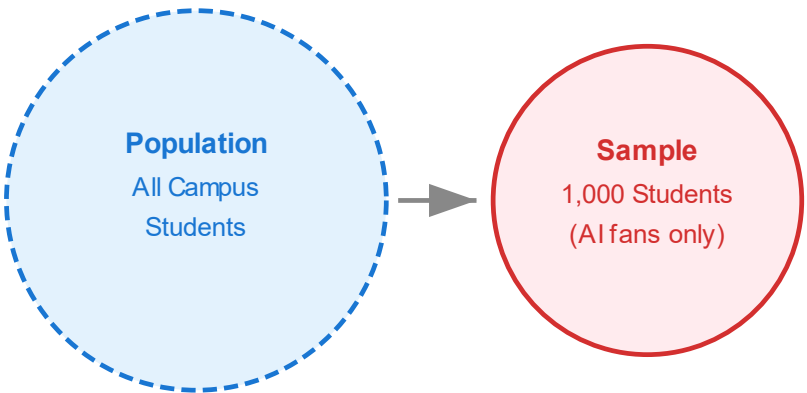
Write T (True) or F (False).

1. A larger-looking bar in a chart always means a larger real difference.
2. A sample can be large and still be biased.
3. Correlation proves causation if the value of r is high.
4. Missing data can affect the validity of conclusions.

1. Misleading Bar Chart — FALSE

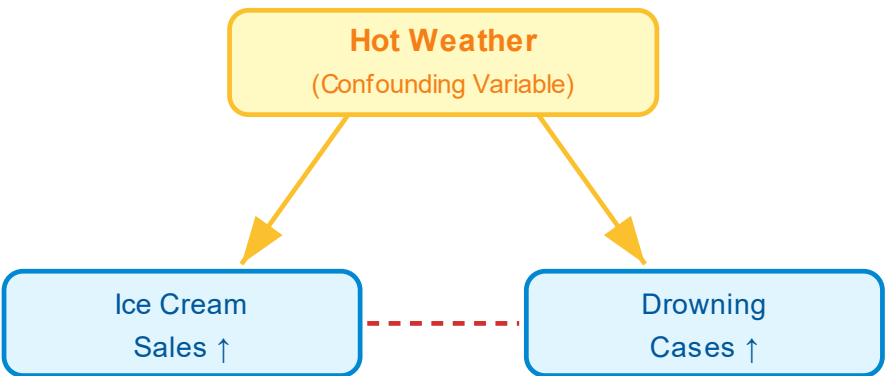


2. Large but Biased Sample — TRUE



Large size ≠ Representative!
Bias comes from HOW you select.

3. Correlation ≠ Causation — FALSE



High r value — but NO causation!
r only shows linear relationship.

4. Missing Data Affects Results — TRUE

Student	Score
Student A	92
Student B	88
Student C	MISSING X
Student D	MISSING X

Missing often = low scorers skipped survey.
→ Average looks higher than reality!
Result: Biased & Less Reliable Conclusion.

Q3. Multiple Choice (Sampling Bias)

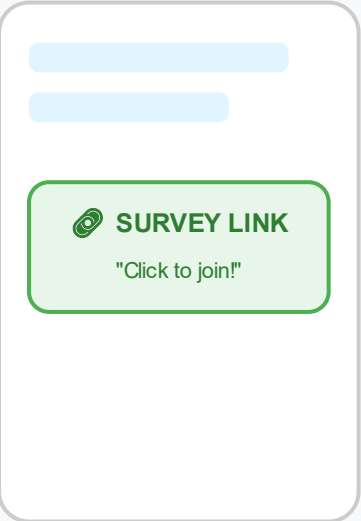
A lecturer posts a survey link in the class group chat and only students who choose to respond are included. This is best described as:

- A. Stratified sampling
- B. Systematic sampling
- C. Voluntary response sampling
- D. Cluster sampling

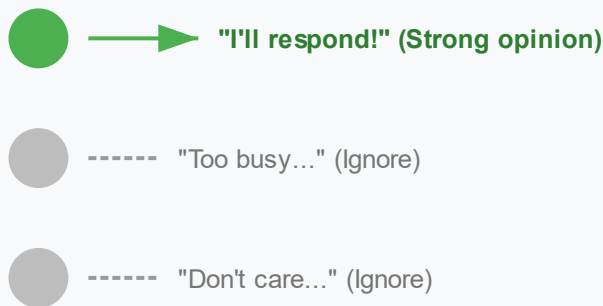
Q3 - Solution

Q3: Voluntary Response Sampling (自愿响应抽样)

1. Scenario: Survey Link in Group Chat



Students decide for themselves:



The Bias (偏差)

- Participants self-select.
- Over-represents those with strong feelings.

NOT REPRESENTATIVE!

Other Options (to be discussed in Q6):

A. Stratified



(By Layers)

B. Systematic

1, 2, [3], 4, 5, [6]
(Every nth)

D. Cluster



(Whole Groups)

C. Voluntary Response (Correct Answer)

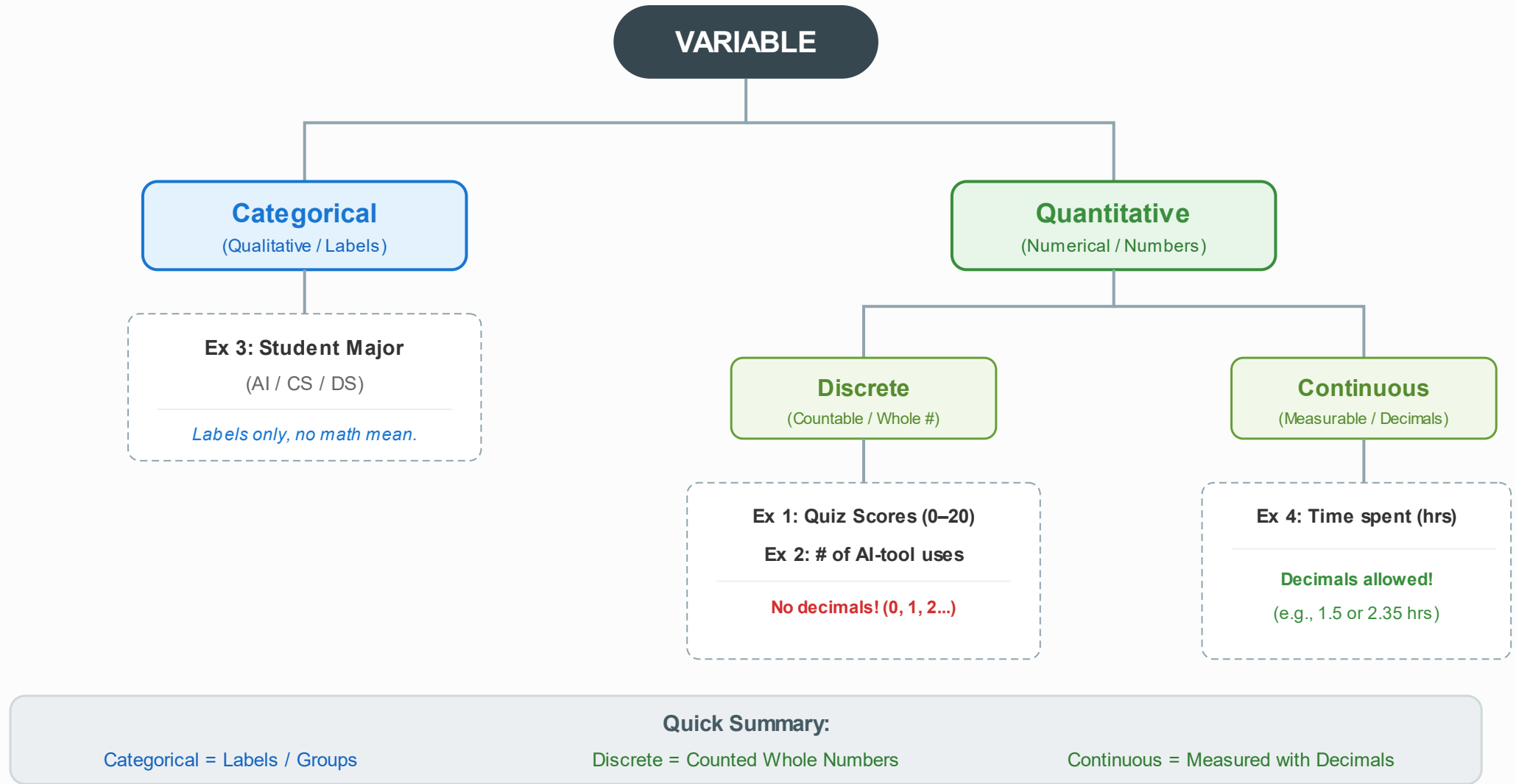
Self-Selection → High Bias

Q4. Variable Type Classification

Classify each variable as Quantitative or Categorical. If quantitative, also classify it as Discrete or Continuous.

1. INT104 quiz score (0–20)
2. Number of AI-tool uses per week
3. Student major (AI / CS / DS)
4. Time spent on revision (hours)

Q4



Q4 - Solution

1. INT104 quiz score (0–20): **Quantitative, Discrete**
2. Number of AI-tool uses per week: **Quantitative, Discrete**
3. Student major (AI / CS / DS): **Categorical**
4. Time spent on revision (hours): **Quantitative, Continuous**

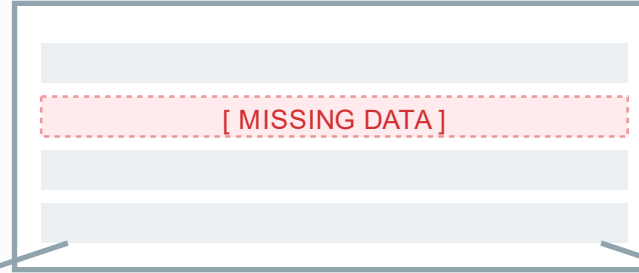
Q5. Missing Data (Comparison Table)

Complete the table with one advantage and one risk for each method.

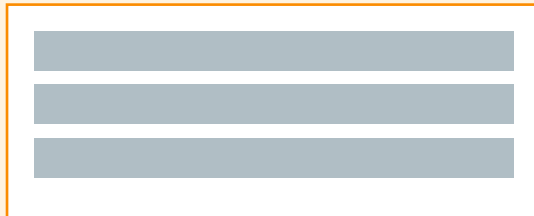
Method	One advantage	One risk
Delete cases		
Impute missing values		

Q5

Original Dataset (with Gaps)



Method A: Delete Cases



← Shorter!

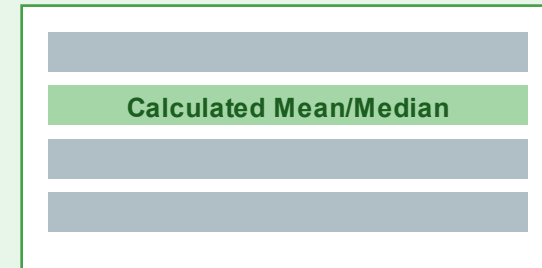
✓ Advantage:

Simple and easy to apply.

✗ Risk:

Reduces sample size (Information loss).
May introduce Bias.

Method B: Impute Values



← Preserved!

✓ Advantage:

Keeps all data / Preserves sample size.

✗ Risk:

Inaccurate values can distort the results.
(Synthetic data vs. Real data)

Q5 - Solution

Q5. Missing Data

Method	One advantage	One risk
Delete cases	Simple and easy to apply	Reduces sample size and may introduce bias
Impute missing values	Keeps more data / preserves sample size	Imputed values may be inaccurate and distort results

Q6. Sampling Methods (Matching)

Match each description to the correct method.

Methods:

- A. Convenience sampling
- B. Systematic sampling
- C. Stratified sampling
- D. Cluster sampling

Descriptions:

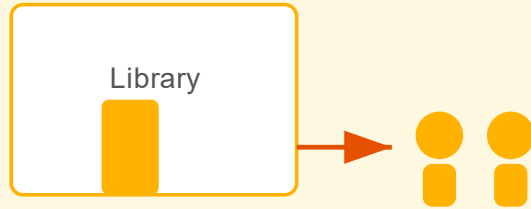
1. Select every 8th student on a list
2. Divide students by year level and sample from each year
3. Survey students leaving the library entrance
4. Randomly choose 2 tutorial groups and survey all students in those groups

Q6

Q6: Four Sampling Methods

A. Convenience Sampling

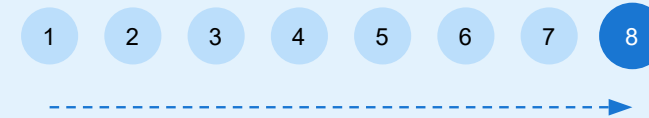
Easiest to Reach



Survey students leaving the library.

B. Systematic Sampling

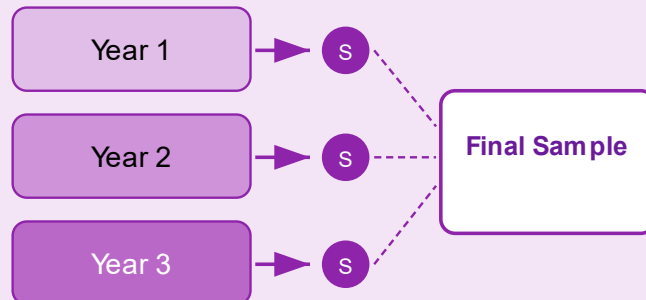
Fixed Interval Selection



Every 8th student selected

C. Stratified Sampling

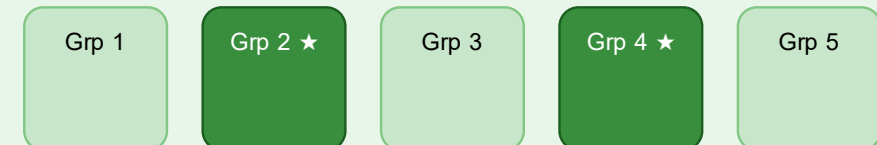
Split into Strata, then Sample



Sample from EVERY layer.

D. Cluster Sampling

Select Whole Groups Randomly



Pick whole groups → Survey ALL in them.

Q6 - Solution

1. Select every 8th student on a list B
2. Divide students by year level and sample from each year C
3. Survey students leaving the library entrance A
4. Randomly choose 2 tutorial groups and survey all students in those groups D

A. Convenience sampling

B. Systematic sampling

C. Stratified sampling

D. Cluster sampling

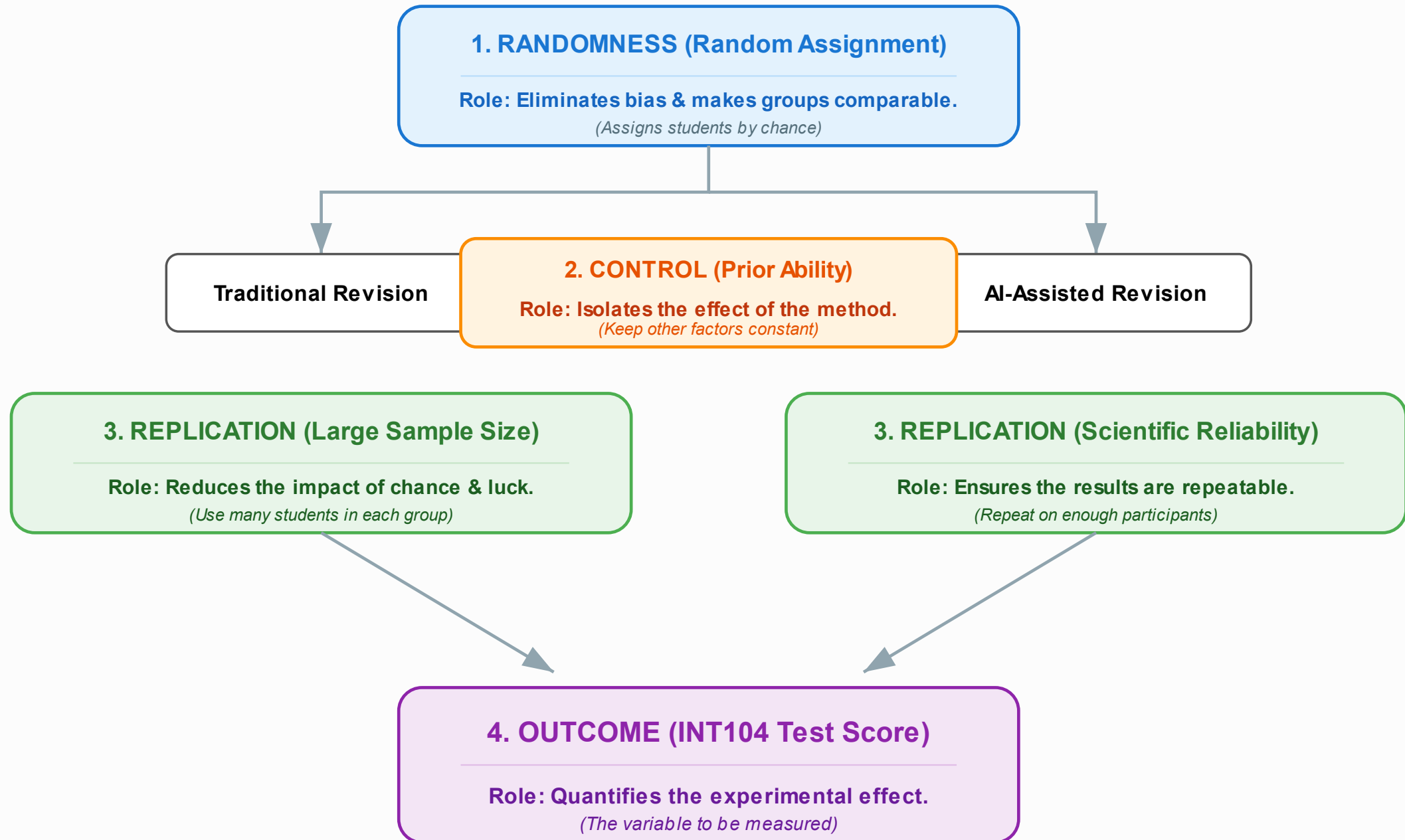
Q7. Experimental Design (Structured Short Answer)

A tutor wants to compare traditional revision and AI-assisted revision for an INT104 test.

Fill in the design plan by giving a brief phrase for each item:

1. **Randomness:** How will students be assigned?
2. **Control of confounders:** Name one confounder to control.
3. **Replication:** What does replication mean here?
4. **Outcome variable:** What will be measured?

Q7: Experimental Design – Core Elements



Q7. Experimental Design

1. **Randomness:** Randomly assign students to the two revision groups.
2. **Control of confounders:** Control prior ability level.
3. **Replication:** Use many students in each group / repeat on enough participants.
4. **Outcome variable:** INT104 test score.

Q8. Histogram Concepts (Short Answer)

Answer the following briefly.

1. What is a histogram used to show?
2. What can happen if the number of bins is too small?
3. What can happen if the number of bins is too large?

1. Function of a Histogram

Shows the **DISTRIBUTION** of a **QUANTITATIVE** variable (Concentration, Shape, Dispersion).

How the Number of Bins Affects the Graph (Same Data)

Q2. Bins are Too SMALL (e.g., 4 bins)

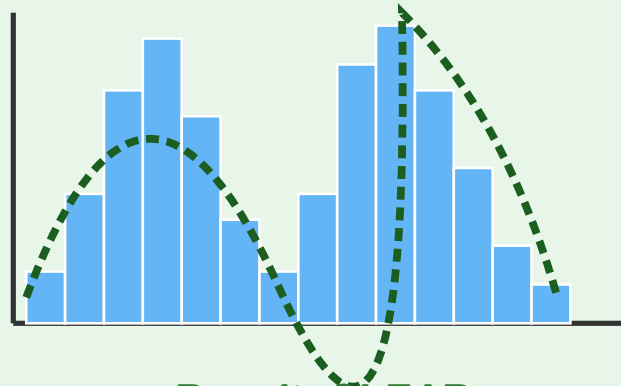


Result: COARSE

Intervals are too wide.

HIDES important patterns!

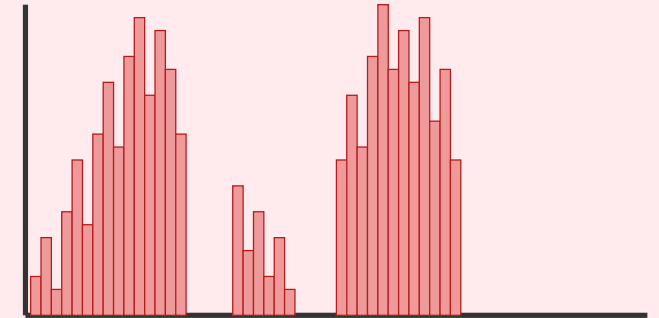
Optimal Bins (e.g., 15 bins)



Result: CLEAR

Reveals the core pattern.
(e.g., Bimodal distribution)

Q3. Bins are Too LARGE (e.g., 50 bins)



Result: NOISY

Intervals are too narrow.
Shows RANDOM fluctuation!

💡 Rule of Thumb: Choose the right number of bins to smooth out noise without losing important patterns.

Q8. Histogram Concepts

1. A histogram is used to show the **distribution of a quantitative variable**.
2. If the number of bins is too small, the graph may be **too coarse and hide important patterns**.
3. If the number of bins is too large, the graph may be **too noisy and show random fluctuation**.

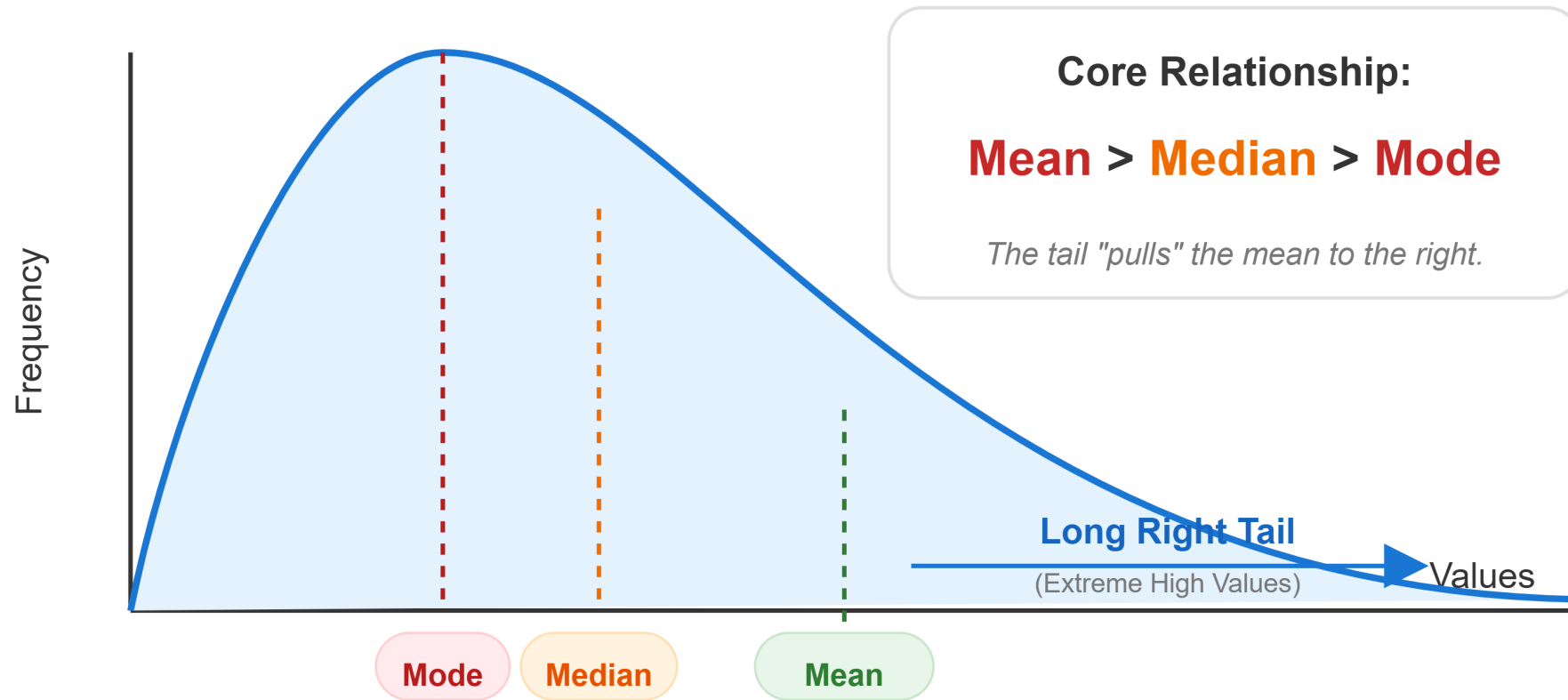
Q9. Distribution Shape (Fill in the blanks)

For a right-skewed distribution, complete the relationship using $>$, $<$, or $\approx$:

Mean____Median____Mode

Then state whether the right tail is longer or shorter than the left tail.

Q9: Right-Skewed (Positive) Distribution



The **Mean** is the average of a data set. It is calculated by adding up all the values and dividing the sum by the total number of observations

The **Median** is the middle value of a data set when the values are arranged in order (from smallest to largest).

The **Mode** is the value that appears most frequently in a data set.

Q9. Distribution Shape

For a right-skewed distribution:

$$\text{Mean} > \text{Median} > \text{Mode}$$

The **right tail is longer** than the left tail.

Q10. Correlation and Significance (Interpretation)

Given:

Sample correlation:

$$r = 0.591$$

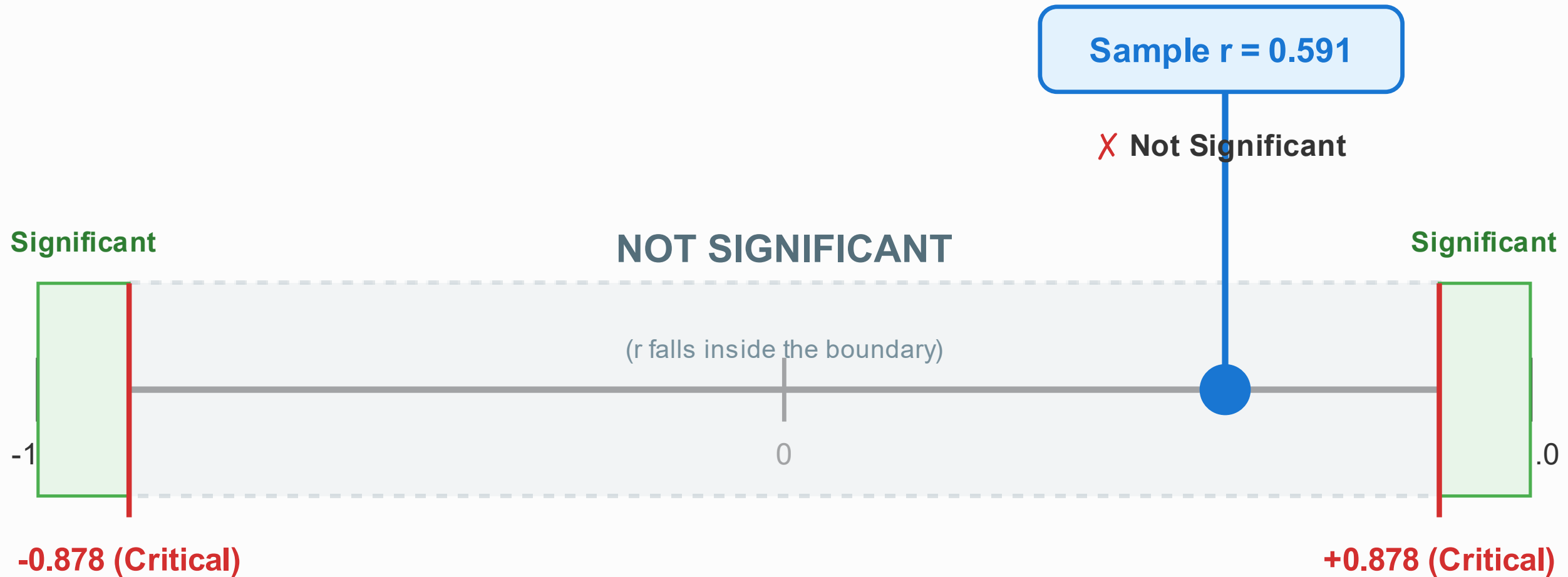
Critical values (class example):

$$\pm 0.878$$

Answer:

1. Is the correlation statistically significant under this criterion? (Yes / No)
2. Complete the reason: It is significant only if r is _____ than or _____ than _____.

Q10: Statistical Significance of Correlation (r)



Decision Rule: Is r greater than $+0.878$ OR less than -0.878 ? → NO.

Q10. Correlation and Significance

Given $r = 0.591$ and critical values ± 0.878 :

1. **No**
2. It is significant only if r is **greater** than $+0.878$ or **less** than -0.878 .

1. Key Factors of Critical Values (Not in exam)

The Critical Value is a dynamic "threshold." It is not fixed because it must adapt to the specific conditions of each study.

Factor	Change	Effect on Critical Value	Logic
Sample Size (n)	Small	Higher Threshold	Small groups have high "noise." You need a huge result to prove it's not luck.
	Large	Lower Threshold	Large groups reduce "noise." Even a small signal can be trusted.
Strictness (α)	Strict (0.01)	Higher Threshold	If you want to be 99% sure, you set a very difficult "pass mark."
	Standard (0.05)	Lower Threshold	A 5% error rate is more lenient, making the "pass mark" easier.

2. The Logic: "Signal" vs. "Noise"

You can simplify statistical testing into a battle between two forces:

- **Signal:** Your Correlation Coefficient (r). This represents the actual relationship you found.
- **Noise:** Random error. The smaller the sample size, the louder the noise.

The Rule: To be "**Statistically Significant**," your Signal must be strong enough to rise above the Noise (the Critical Value).

3. Step-by-Step Decision Process

When solving problems, follow this logic:

1. Identify the Threshold: Look for the Critical Value provided in the problem (e.g., 0.878).

2. Check Your Result: Look at your calculated $|r|$ (e.g., 0.591).

3. Compare:

1. If $|r| > \text{Critical Value} \rightarrow$ **Significant**. The signal beat the noise.
2. If $|r| < \text{Critical Value} \rightarrow$ **Not Significant**. The signal was too weak.

Conclusion for your case: Because 0.591 is **smaller** than 0.878, the result is **not significant**. The "noise" was too high for such a small sample.

Q11. Boxplot Practice (Calculation + Interpretation)

A dataset has the following five-number summary:

$$\text{Minimum} = 10$$

$$Q_1 = 25$$

$$\text{Median} = 35$$

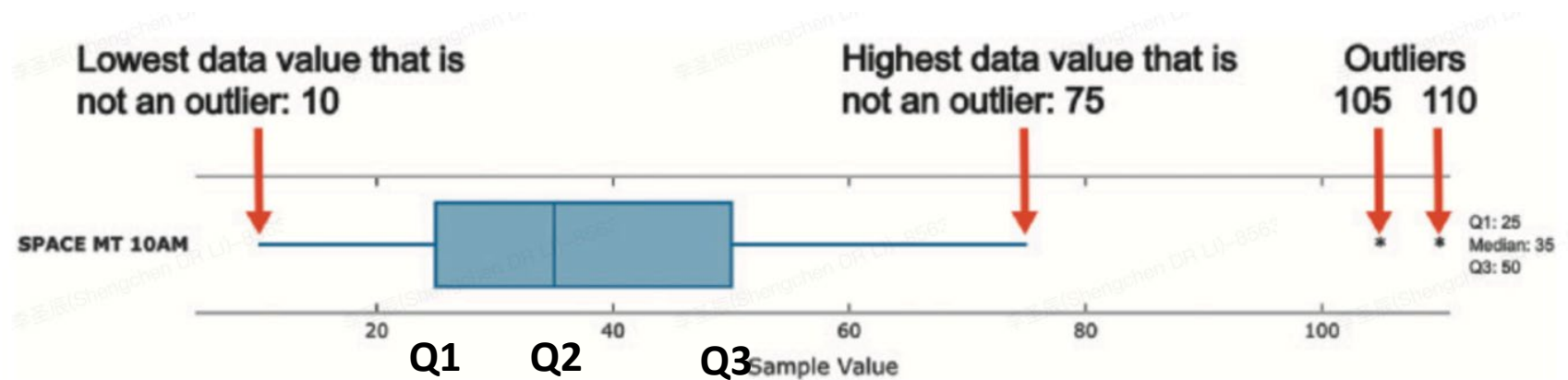
$$Q_3 = 50$$

$$\text{Maximum} = 110$$

A modified boxplot shows 105 and 110 as outliers; the highest non-outlier value is 75.

Answer:

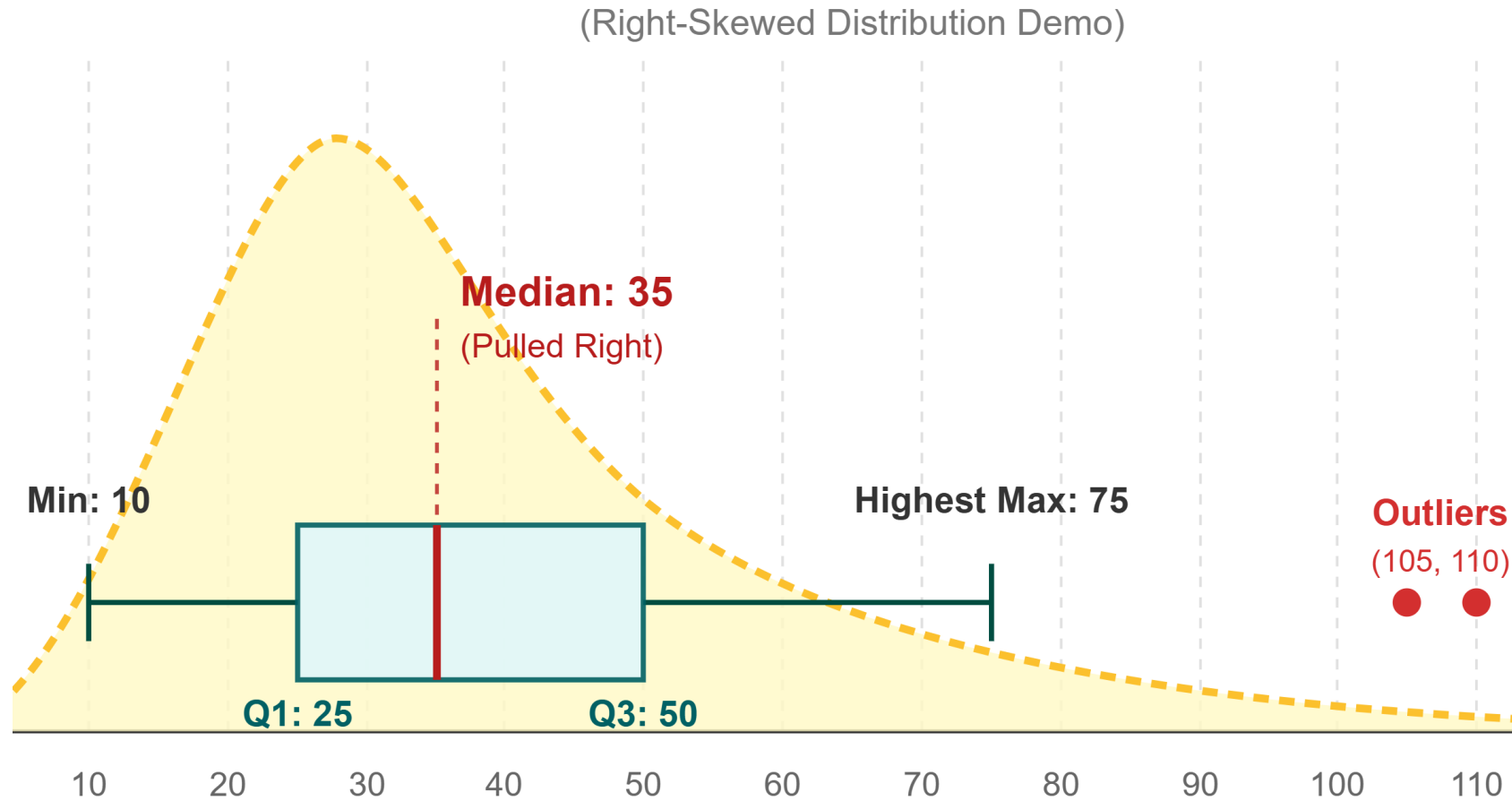
1. Compute the IQR.
2. State the interval containing the middle 50% of observations.
3. Based on the high-end outliers, the distribution is more likely to be:
 - A. Left-skewed
 - B. Right-skewed
 - C. Symmetric



Boxplot Part	Corresponds To	Meaning
Left edge of box	Q1 = 25	Bottom 25% boundary, 25% of data falls below this value
Line inside box	Median (Q2) = 35	Exact middle of the data, 50% of data falls below this value
Right edge of box	Q3 = 50	Top 25% boundary, 75% of data falls below this value
Width of the box	IQR = 25	Spread of the middle 50%
Dots beyond whiskers	Outliers (105, 110)	Extreme values far from the box

Q1, Q2, Q3 are the three dividing points that split your data into four equal parts.

IQR is simply the width of the middle section — it tells you how spread out the core data is, without being affected by extreme values.



1. IQR stands for Interquartile Range, which is a key index of boxplots: $IQR = Q3 - Q1$
2. The interval containing the middle 50% of observations

Q11. Boxplot Practice

Given $Q_1 = 25$, $Q_3 = 50$, and high-end outliers at 105 and 110.

1.

$$\text{IQR} = Q_3 - Q_1 = 50 - 25 = 25$$

2. The middle 50% of observations lie in the interval:

$$[25, 50]$$

3. **B. Right-skewed**